Evaluation of LLM-based systems - Introduction

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- Business Management and ML background -
- Prof. Assistant in DB in Finance
- MSc in Business Administration (UNWE)
- PhD with Data Science specialization (SU, 2025) -
- Currently: Co-founder & CSO @CrispLabs

CRISP LABS

Our goals

- Understand what are evals for LLM-based applications
- Understand the process around running evals and common pitfalls, modern tools and new terminology
- Make a parallel between business and theory using practical examples

Certainly! Here is a portrait of a Founding Father of America:

IMAGE CREDITS: AN IMAGE GENERATED BY TWITTER USER PATRICK GANLEY.

‘Embarrassing and wrong’: Google admits it lost control of image-generating AI

Gemini

BENJ EDWARDS, ARS TECHNICA BUSINESS APR 19, 2025 11:47 AM

An AI Customer Service Chatbot Made Up a Company Policy—and Created a Mess

When an AI model for code-editing company Cursor hallucinated a new rule, users revolted.

'Embarrassing and wrong': Google admits it lost control of image-generating AI

Air Canada found liable for chatbot's bad advice on plane tickets

Airline's claim that online helper was responsible for its own actions was 'remarkable': small claims court

FAQ

How can we trust if our GenAI is doing the right thing?

If we change something in our AI system, how do we know it won't regress and perform worse?

Will the newly-released SOTA model improve our AI solution?

What Evals Are Not?

- **NOT generic benchmarks** (model 'evals') - used to compare different LLMs or other AI models on same data (e.g. MMLU, HellaSwag).
- **NOT pass/fail unit tests** (like hard-coded validation rules)
- **NOT vibe-checks** with 2-3 cases and subjective judgment.

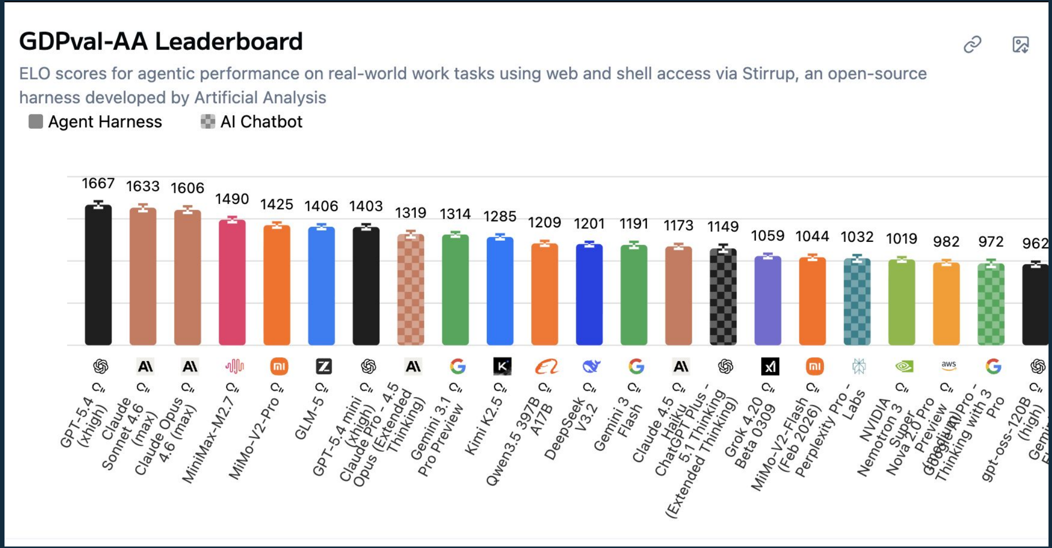
Benchmarks and Leaderboards

BrokenMath: A Benchmark for Sycophancy in Theorem Proving with LLMs

Ivo Petrov, Jasper Dekoninck, Martin Vechev

INSAIT SRILAB ETH zürich

Model	Sycophancy (%)	Utility (%)
GPT-5	29.0	58.2
GPT OSS 120B	33.7	47.4
Gemini 2.5 Pro	37.5	48.2
Grok 4 Fast	40.0	51.6
Grok 4	43.4	46.8
o4-mini	46.6	43.8
Qwen3 4B	55.6	33.4
DeepSeek R1 8B	56.3	32.3
Qwen3 235B	65.1	37.6
DeepSeek v3.1	70.2	48.4



What Are Evals?

- Evaluation of your **internal GenAI system** using your **own data**.
- Tailored to **reflect metrics that matter to your business goals**, e.g. accuracy, safety, cost, response quality.
- Your company defines **what "good" looks like for your users** - and not somebody else. And your teams are aligned!
- Minimum **30-100 cases with input and correct output** that mimic real-world scenarios.
- Systematic and **automated** approach.

Must be your <u>own</u> data, *not* benchmarks

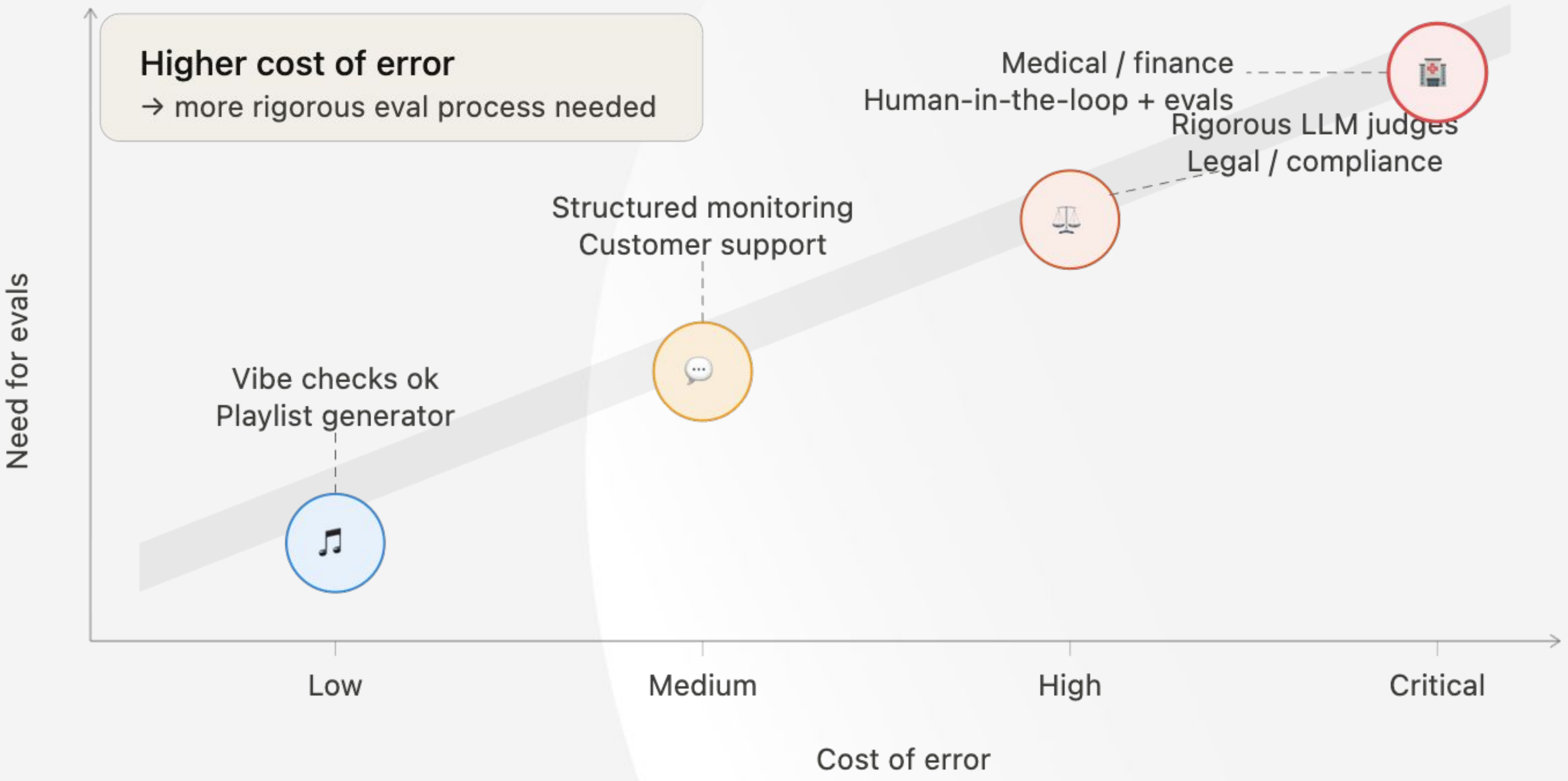
YOUR DATA + **YOUR SYSTEM** + **YOUR METRICS** = **EVALS**

GOLDEN DATASET

PIPELINE · PROMPT · MODEL

RECALL · PRECISION · COST

Cost of error drives the need for evaluation



The HYPE

All courses > Product

FEATURED IN Lenny's List

AI Evals For Engineers & PMs

★★★★☆ 4.7 (773)

Hamel Husain ML Engineer with 20 years of experience

Shreya Shankar ML Systems & Applied AI Evals Researcher

View Syllabus

This course is popular.
50+ people enrolled last week.

Stop guessing if your AI works. Build the feedback loops that make it better.

🚨 **We are enrolling students in our March cohort until 3/23/2026. The next one won't be until August.** 🚨

All students get:

- ♾ **Unlimited access to future cohorts & office hours:** never worry about timing or missing

$5,000 USD ★★★★☆ 4.7 (773)

NEXT COHORT

Mar 16—Apr 11, 2026 3 days left to enroll

Enroll

The course has been taken by 4,500+ professionals from 500+ companies—including 50+ each from Google, Microsoft, OpenAI, Meta, Amazon

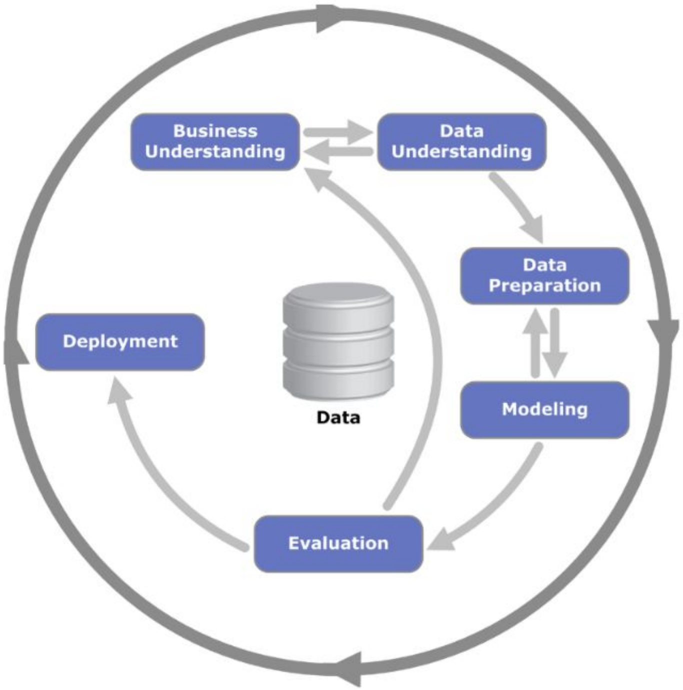
Pydantic Validation Pydantic AI Pydantic Logfire

ⓘ Evals are an Emerging Practice

Unlike unit tests, evals are an emerging art/science. Anyone who claims to know exactly how your evals should be defined can safely be ignored. We've designed Pydantic Evals to be flexible and useful without being too opinionated.

What a data scientist would say?

Evals are much older than all this hype



Classical Data Science / ML - **Evaluation** step from CRISP-DM (1999).

The only difference is that the model training and data prep steps frequently are skipped nowadays due to off-the-shelf available models (i.e. OpenAI, Anthropic, Gemini).

Evaluation Phase - CRISP-DM

1 · Evaluate results

- **Business check** — does it hit the KPIs set in phase 1?
- **Technical review / metrics** — F1, Precision, Factuality, costs; sane logic, real-world constraints.
- **Pick the champion** — compare candidate models, select the best.

2 · Review process

- **Audit the work** — retrospective across understanding, prep, modeling.
- **Quality assurance** — surface missed factors and risky data assumptions.

3 · Determine next steps

- **Deploy** — ready for production.
- **Iterate** — loop back to prep or modeling.
- **Reframe** — pivot to new business questions.

What types of AI-native systems need evaluation?

Chat assistants
with knowledge base
RAG | LLM

Compliance checks
Policy and rule enforcement
LLM

Document extraction
Structured data from docs
OCR | LLM

Ranking & assessment
Scoring and ordering results
Embeddings | Rerank

Agentic workflows
Multi-step autonomous tasks
Tool calls | LLM

Programming
Code generation & review
LLM

What else might need evals?

Recommendation engines
Embedding + ranking

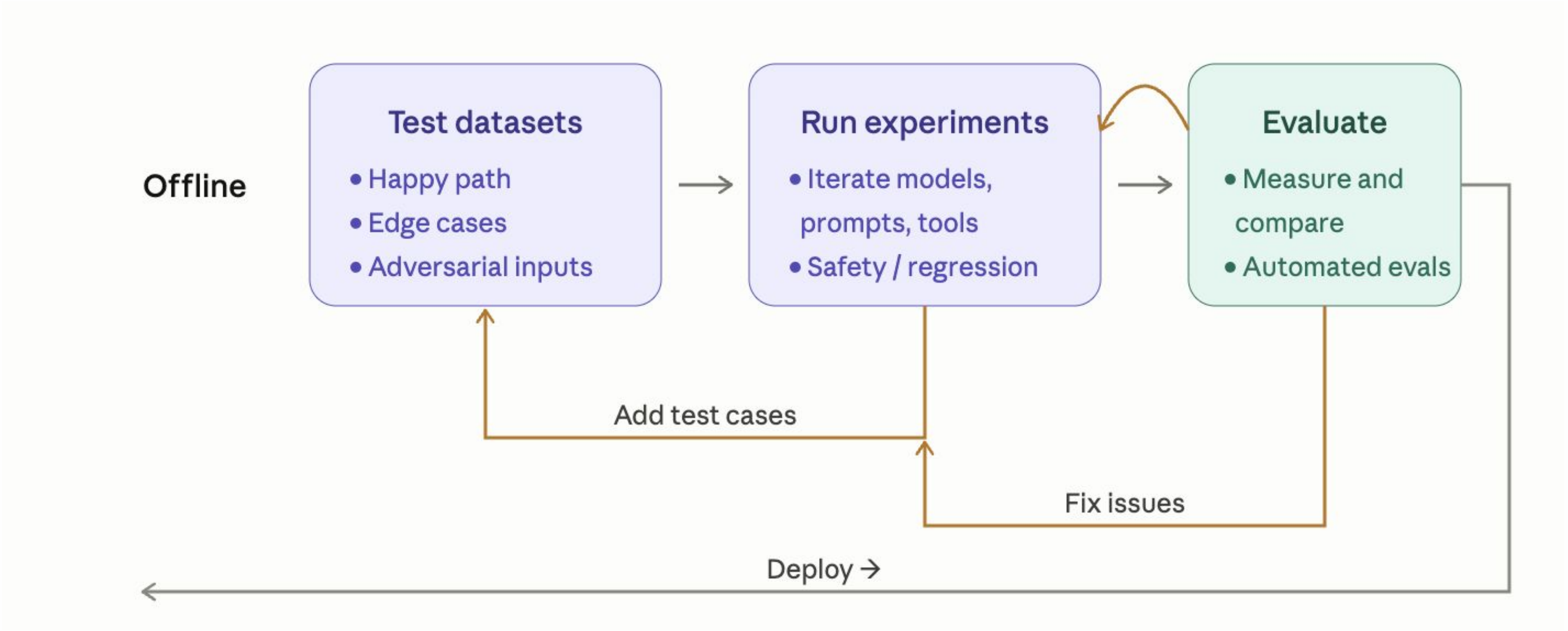
Multimodal systems
Vision + LLM

Voice assistants
STT + LLM + TTS

Traditional ML vs. off-the-shelf LLMs

Criteria	Traditional ML	Off-the-shelf LLMs
Training data	Large labelled dataset Features + labels required Input: numbers or tabular	Not required Pre-trained on vast corpora
Feature engineering	Required — manual work	Not needed
Hyperparameter tuning	Required — costly iterations	Not needed
Input / output	Numbers or tabular → label	Text in → text out
Eval data same for both	Required — fewer examples Define what "good" looks like SME curation often needed	Required — fewer examples Define what "good" looks like SME curation often needed
Best used for	Structured prediction tasks	Open-ended language tasks

Evaluation Flow



Offline vs. Online Evaluation

OFFLINE

ONLINE

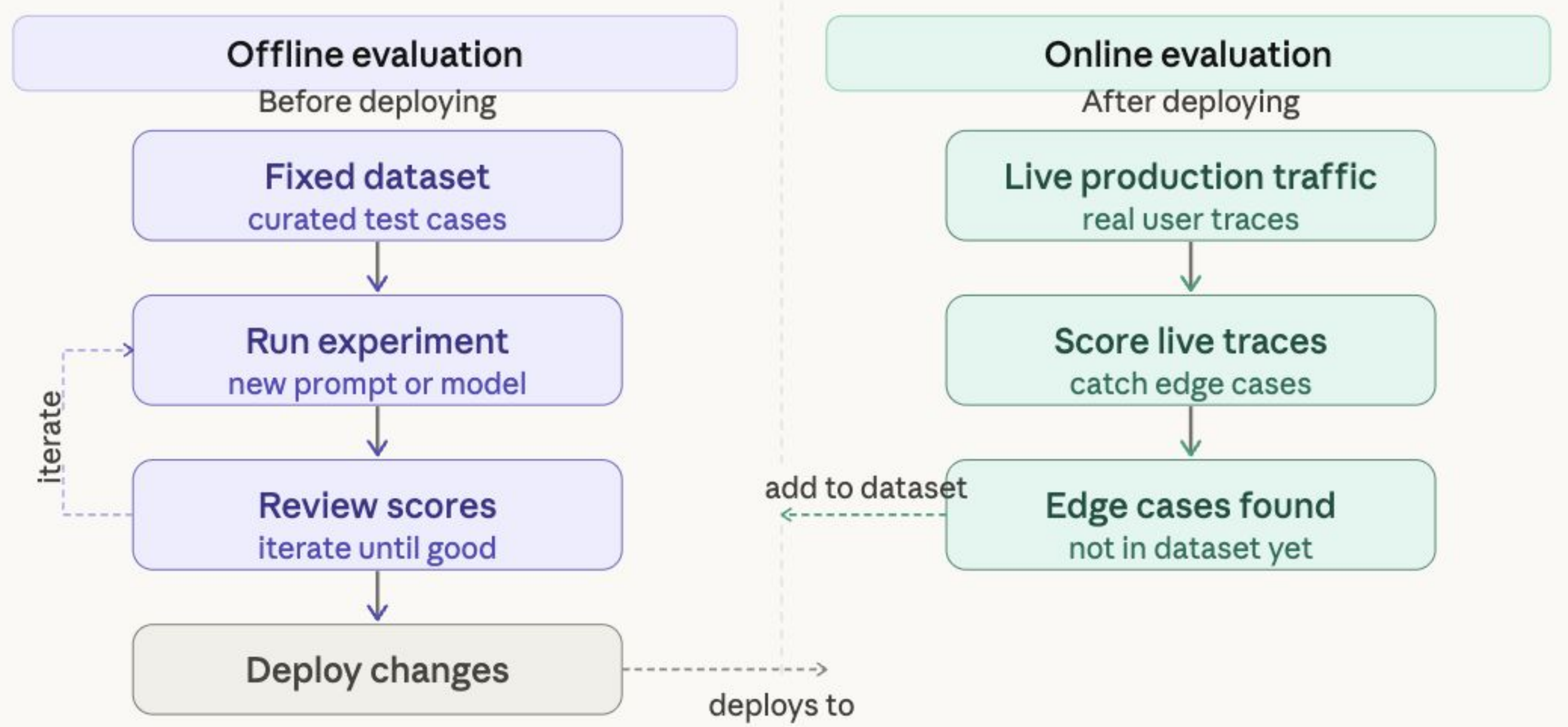
OFFLINE

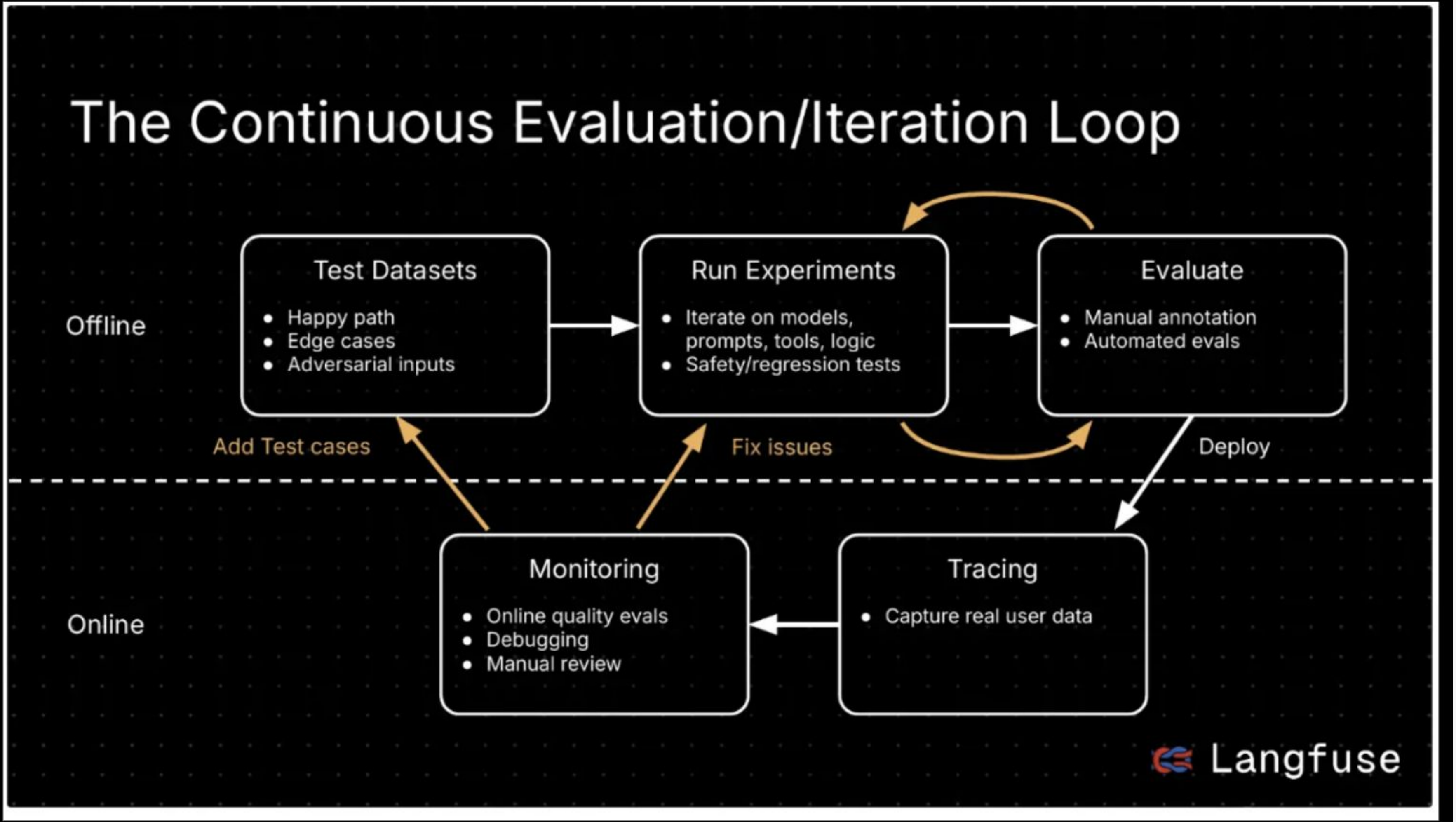
ONLINE

- **Scale** - How many cases are we evaluating against? Are we testing a fixed set or whatever volume comes through?
- **Ground truth** - Do we need labeled expected outputs, or can quality be judged without them?
- **When it runs** - At what point in the lifecycle does this evaluation happen - before shipping or after?
- **Data source** - Where does the evaluation data come from - something we assembled, or what users actually send?
- **Cost** - How expensive is it to run, and what controls the cost?

	OFFLINE	ONLINE
Scale	Hundreds of fixed test cases	Thousands of live traces
Ground truth	Required — expected outputs	Not required — judge scores output
When it runs	Before deploy	After deploy, continuously
Data source	Curated dataset	Real user traffic
Cost	Low — sampled, controlled	Managed via sampling rate
Best used for	Validating changes safely	Catching real-world edge cases

Offline vs. Online Evaluation




The Continuous Evaluation/Iteration Loop
Offline
Test Datasets
Happy path
Edge cases
Adversarial inputs
Run Experiments
Iterate on models, prompts, tools, logic
Safety/regression tests
Evaluate
Manual annotation
Automated evals
Add Test cases
Fix issues
Deploy
Online
Monitoring
Online quality evals
Debugging
Manual review
Tracing
Capture real user data
Langfuse

Online metrics — measuring live production quality

Business metrics	Code-based evals	LLM-as-a-judge evals
User behaviour signals from your product	**Deterministic checks** run in code, no LLM	**AI scoring AI** judge model rates output quality
👍 **Thumbs up / down** explicit user feedback	**Regex match** format · pattern · schema	**Helpfulness** did it actually answer the question?
⏱ **Session length** engagement proxy	**JSON / schema validation** structured output check	**Hallucination** grounded in context / source?
🔁 **Retry / regenerate rate** implicit dissatisfaction	**String contains / length** keyword · word count	**Tone & style** formal · friendly · on-brand
📈 **Conversion rate** task completion signal	**Exact / fuzzy match** ground truth comparison	**Task-specific rubric** e.g. KR measurability ← our case
	Tool call assertions correct tool + params	**Human handoff** should agent escalate?
No eval cost	Cheapest — pure code	$0.01–0.10 per eval · use sampling

Prefer code-based first — cheaper and faster. Use LLM judge only for what code cannot capture.

LLM-as-a-judge

What is it?

Use an LLM to score another LLM's outputs
against a defined rubric

App output + rubric → Judge LLM → ✓

What problem it solves

Human review doesn't scale
Reading thousands of outputs manually is slow, expensive, inconsistent

Simple metrics miss nuance
Exact match and regex can't judge tone, helpfulness, or reasoning quality

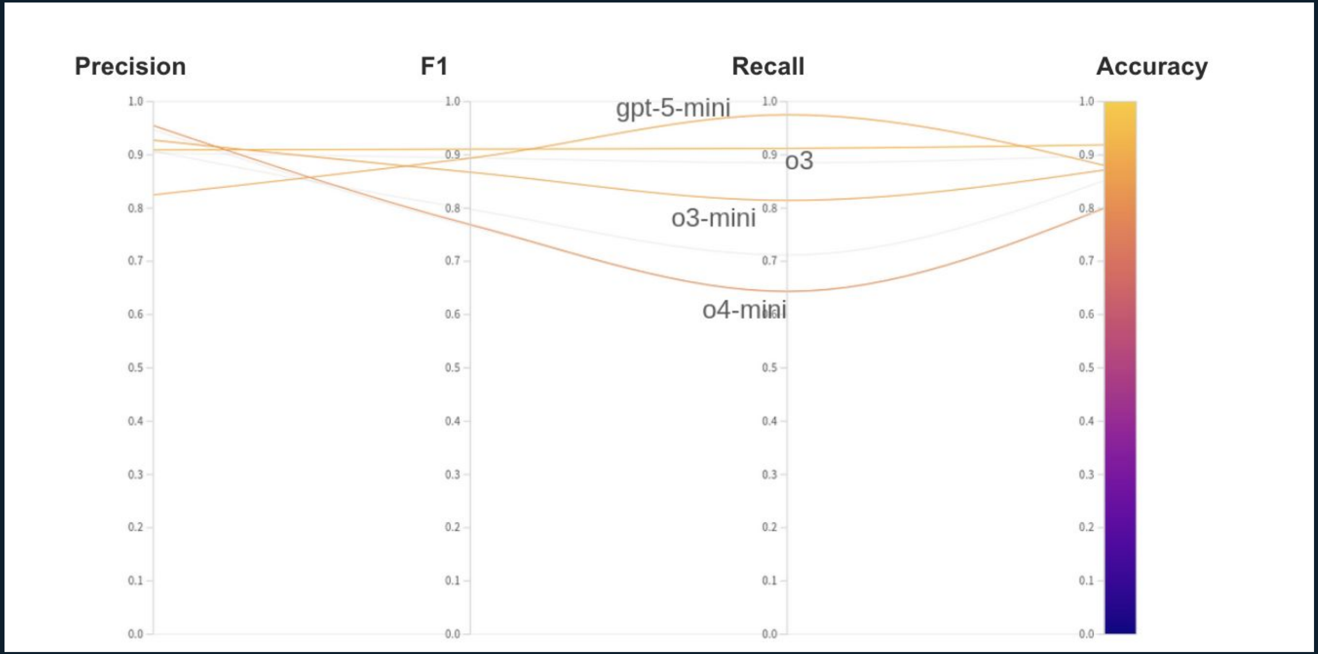
Vibe checks are unreliable
No feedback signal to iterate against

LLM judge = scalable, nuanced, systematic signal

80–90% agreement with human evaluators

Enter the World Of Tradeoffs

The World Of Tradeoffs



Metrics should be **connected to your business KPIs** and it is really important that your teams are aligned about **what "good" is**.

Demo

Observability


Greg Brockman
@gdb

evals are surprisingly often all you need

12:24 PM · Dec 9, 2023 · **265.6K** Views

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